

GVARV, Norway

WMO: 014700

Lat: 59.383N

Lon: 9.201E

Elev: 305

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	8.2	-3.5	4.6	25.2	1.7	6.1	16.0	15.7	35.0	13.8	35.6	4.4	50	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	79.8	63.4	76.2	60.9	72.9	59.1	65.1	76.2	63.3	72.3	61.7	69.4	4.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	81.4	67.8	59.7	77.3	66.3	58.2	73.3	64.9	30.1	76.1	28.8	72.6	27.7	69.3	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.6	(b) 11.5	(c) 9.8	(e) 1.7	(f) 85.4	(g) 4.6	(h) 2.9	(i) -1.7	(j) 87.5	(k) -4.4	(l) 89.3	(m) -7.0	(n) 90.9	(o) -10.4	(p) 93.0		
(4) 13.6	(5) 11.5	(6) 9.8	(7) 1.7	(8) 85.4	(9) 4.6	(10) 2.9	(11) -1.7	(12) 87.5	(13) -4.4	(14) 89.3	(15) -7.0	(16) 90.9	(17) -10.4	(18) 93.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	26.5	27.9	35.4	43.8	51.2	58.6	63.0	60.4	53.3	43.5	35.5	28.1		
	DBStd	14.34	8.28	8.39	7.38	5.88	6.37	4.68	5.00	4.05	4.90	6.06	6.50	9.59		
	HDD50	3418	728	619	455	202	59	2	0	0	24	215	436	678		
	HDD65	7717	1193	1039	919	635	431	200	102	152	350	667	886	1143		
	CDD50	1234	0	0	1	17	96	259	402	321	124	13	1	0		
	CDD65	57	0	0	0	0	3	7	39	8	0	0	0	0		
	CDH74	548	0	0	0	0	34	114	344	56	0	0	0	0		
	CDH80	83	0	0	0	0	3	16	62	2	0	0	0	0		
	WSAvg	4.3	4.5	4.5	4.8	4.8	4.7	4.4	3.9	3.6	4.0	4.0	4.4	4.4		
	PrecAvg	37.6	3.1	1.9	1.7	2.1	2.8	3.3	3.7	4.4	3.9	4.1	3.8	2.5		
(15) Precipitation	PrecMax	56.5	7.2	5.9	4.7	5.1	8.0	6.7	9.1	7.3	10.8	10.0	16.5	4.6		
	PrecMin	29.2	0.5	0.4	0.1	0.3	0.3	0.2	0.8	0.3	1.5	0.6	0.6	0.4		
	PrecStd	6.1	1.7	1.3	1.1	1.3	1.7	1.7	2.1	1.9	2.0	2.2	3.1	1.1		
	PrecStd	6.1	1.7	1.3	1.1	1.3	1.7	1.7	2.1	1.9	2.0	2.2	3.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	46.1	48.3	60.2	68.5	78.5	82.4	85.4	79.1	71.3	60.8	54.2	50.4			
	0.4% MCWB	41.9	40.4	46.8	52.6	60.4	59.5	66.1	64.4	59.2	52.6	48.7	45.8			
	2% DB	42.5	44.3	54.7	64.4	72.7	77.1	81.5	75.0	68.2	57.6	50.8	46.3			
	2% MCWB	38.5	38.0	45.3	50.5	56.1	57.9	65.4	61.3	57.6	51.8	47.9	42.8			
	5% DB	40.2	41.5	50.0	59.7	68.1	73.3	78.0	71.9	65.0	55.3	48.0	43.7			
	5% MCWB	36.8	36.8	41.5	48.4	54.4	56.6	63.6	59.8	55.7	50.8	45.5	40.6			
	10% DB	37.5	38.8	46.1	55.2	63.4	69.7	74.5	69.4	61.9	53.0	45.3	40.8			
	10% MCWB	34.9	35.8	38.9	45.2	51.9	55.4	61.4	58.8	54.6	49.2	43.2	38.3			
	0.4% WB	42.8	42.1	49.1	54.4	62.0	63.3	68.0	65.9	62.0	55.1	51.0	46.6			
	0.4% MCDB	45.3	46.1	58.4	65.9	74.7	75.5	81.2	75.8	68.7	58.2	52.1	50.0			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	39.9	39.5	45.5	51.3	58.9	61.2	66.2	63.6	59.5	53.5	48.1	43.4			
	2% MCDB	41.9	42.9	54.1	60.9	68.3	72.2	78.9	71.1	64.3	56.0	49.5	45.7			
	5% WB	37.3	37.6	42.5	49.1	55.7	59.3	64.6	62.0	57.9	51.8	45.8	40.9			
	5% MCDB	39.6	40.6	48.9	57.9	65.0	68.5	75.8	68.7	62.5	54.2	47.9	43.1			
	10% WB	34.9	35.8	39.6	46.9	53.2	57.7	62.9	60.5	56.4	49.8	43.3	38.1			
	10% MCDB	37.0	38.4	44.7	54.4	61.9	66.4	71.6	66.6	60.8	52.5	45.1	40.4			
(35) Mean Daily Temperature Range	MDBR	8.9	11.0	16.0	17.7	17.9	18.6	17.5	17.0	15.9	12.4	8.8	8.8			
	5% DB	11.2	15.7	22.3	25.7	24.8	25.1	23.8	20.9	20.5	15.3	13.1	12.1			
	5% WB	9.3	10.6	14.1	14.2	12.1	12.6	12.2	10.9	11.4	10.3	10.6	9.9			
	MCDBR	10.7	12.6	21.1	23.2	21.4	20.8	21.7	17.4	16.3	12.6	11.8	11.8			
	MCWBR	9.4	9.2	14.1	13.6	11.2	11.3	11.9	10.2	10.3	9.8	10.5	10.4			
	MCWBR	9.4	9.2	14.1	13.6	11.2	11.3	11.9	10.2	10.3	9.8	10.5	10.4			
(40) Clear-Sky Solar Irradiance	taub	0.257	0.295	0.321	0.364	0.361	0.350	0.371	0.365	0.336	0.321	0.278	0.232			
	taud	2.218	2.297	2.322	2.323	2.425	2.476	2.435	2.444	2.520	2.481	2.312	2.144			
	Ebn at Noon	176	229	259	266	276	280	271	263	255	221	168	140			
	Edh at Noon	13	21	28	34	33	31	32	30	23	18	12	10			
	RadAvg	107	314	796	1269	1553	1782	1563	1233	804	375	140	71			
(45) All-Sky Solar Radiation	RadStd	22	67	69	181	192	143	200	91	112	71	37	8			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	+2.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page